

Game Theory and Multi-Agent DRL Based Anti-Jamming Transmission for Integrated Air–Ground Network

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Abstract—This paper proposes an anti-jamming transmission algorithm based on game theory and multi-agent deep reinforcement learning (MADRL) for the integrated air-ground network. For uplink process, we propose UAV deployment schemes based on congestion game model and dynamic networking schemes based on coalition game, aiming to counteract malicious jamming from ground and air jammers, effectively enhancing the anti-jamming transmission capability of air-ground networks. For downlink process, to address the joint trajectory and power optimization problem, a partially observable Markov decision process (POMDP) framework is utilized, follows a centralized training and distributed execution framework. During the centralized training process, experiences of each agent interacting with the environment are stored in an experience replay pool and then used to train the soft actor-critic network. This process is conducted on the high altitude platforms (HAP) to alleviate the burden of unmanned aerial vehicle (UAV). During the distributed execution process, each UAV uses the trained actor network to output actions based on observations and adjust its flight position and transmission power for joint service provision. To update the parameters of the soft actor-critic network, an improved proximal policy optimization (PPO) algorithm is proposed. Simulation results demonstrate that the proposed method outperforms traditional algorithms in terms of achieving higher system achievable sum rate, lower power consumption, and faster convergence speed.

Index Terms—Air-ground network, anti-jamming, deep reinforcement learning, game theory, multi-agent.

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I. INTRODUCTION

THE new generation of mobile communication technologies significantly improve frequency reuse and high-capacity transmission. However, in areas with complex terrain, the node distribution of communication system is sparse, making it challenging to achieve ultra-dense networking and resulting in significant challenges to regional coverage and “mountain connectivity”. Unmanned aerial vehicles (UAVs) and high altitude platforms (HAPs) have the characteristics of flexible deployment and wide coverage, providing a new approach to effectively enhance communication coverage capabilities, playing a crucial role in various applications including emergency rescue, environmental monitoring, data collection, and user communication.

A. Related Works

Due to the flexibility and controllability, UAVs are well-suited for providing reliable wireless communication in specific scenarios [1], [2]. In emergency situations where ground base stations are unable to provide service, UAVs can serve as aerial base stations [3]. Additionally, UAVs are widely used in the Internet of Things (IoT) to collect data for device applications [4]. UAVs can also act as relay nodes to establish reliable connections with remote nodes, withstand malicious jamming and eavesdropping, and ensure secure transmission within UAV relay networks [5], [6]. In addition, the research on anti-jamming issues in UAV networks is also a major research focus [7], [8], [9], where game theory and convex optimization theory are widely applied in the development of anti-jamming schemes for UAV communication. Ref. [7] considers UAVs as relay nodes and constructs a Stackelberg game model from the perspective of game theory to study how the integrated air-to-ground network can resist malicious jamming under imperfect channel state information (CSI) conditions. Ref. [8] applies alternating optimization and successive convex approximation theory to transform non-convex problems, optimizing the trajectories and power allocation of UAVs to maximize the average secrecy rate of the system within a specific time period. Ref. [9] investigated the physical layer security issues of UAV networks, enabling the UAV base station to transmit confidential information to multiple information receivers through cooperative interference in the presence of multiple eavesdroppers, thus improving system performance gains.

In the presence of intelligent jamming patterns, the jammer has the capability to dynamically adjust its jamming strategy by predicting the transmission strategies of legitimate nodes [10]. Conventional fixed anti-jamming solutions are proved to be insufficient in achieving substantial anti-jamming objectives. Currently, there are various researches that focus on the application of intelligent algorithms, such as reinforcement learning (RL) and deep reinforcement learning (DRL), in anti-jamming transmission [11], [12], [13], [14], [15], [16], [17]. Ref. [12] proposed a knowledge-based RL method for UAV networks that were susceptible to jamming from intelligent jammers. Ref. [13] investigated the secure transmission problem in UAV networks under the presence of eavesdropping nodes and proposes a multi-agent deep reinforcement learning (MADRL) algorithm. Ref. [14] obtained the strategy of legitimate communication link nodes and optimized the power and trajectory of the legitimate UAV. The limitation of this algorithm is that it only considers two UAV agents, without considering multi-agent collaborative jamming resistance. Ref. [15] proposed a cooperative multi-agent hierarchical Q -learning anti-jamming transmission algorithm of joint channel and power allocation in UAV networks, which obtained superior performance compared to traditional multi-agent Q -learning algorithms. In [16], an intelligent anti-jamming communication approach for wireless sensor networks was investigated. A random game framework was introduced to model and analyze the challenges posed by multi-user jamming scenarios. On the other hand, [17] presented a UAV swarm communication scheme based on multi-agent RL. This scheme aimed to optimize relay selection and power allocation, thereby enhancing the overall anti-jamming performance.

In-depth research has been conducted on the UAV deployment and networking optimization issues in [18], [19], [20], [21], [22]. Ref. [18] employs particle swarm optimization algorithm to optimize the 3D deployment of multiple UAVs. Ref. [19] studies the joint optimization problem of UAV beamforming, rate allocation, and position deployment assisted by location information, using Big-M method, penalty function method, semidefinite programming, and successive convex approximation to iteratively optimize and converge sub-problems. Ref. [20] combines deep reinforcement learning and game theory to study the power allocation and three-dimensional deployment problems of UAV-base station downlink systems, proposing a price-based proximal policy optimization algorithm to explore optimal strategies, significantly improving system throughput and energy efficiency. Ref. [21] investigates the resource allocation and position deployment problems in multi-UAV communication systems, helping alleviate inter-cell interference affecting system performance. Ref. [22] considers covert communication scenarios between multiple UAV base stations and ground users, improving the minimum value of the average rate lower bound for all users through joint optimization of user association, bandwidth allocation, UAV transmission power control, and 3D deployment.

B. Motivations

The aforementioned research [5], [6], [7], [8], [9] only considers scenarios with a single or a small number of UAVs, and the

issue of anti-jamming in large-scale UAV scenarios remains to be studied. In addition, there are still three issues in anti-jamming research based on game theory and convex optimization theory applied in aforementioned research. Firstly, most game theory based algorithms require the base station to know the accurate jamming model [10], which is difficult to obtain in actual adversarial environments. Secondly, traditional convex optimization methods are limited to achieving local optimal solutions. As the model becomes more complex and the number of optimized variables increases, non-convex problems become more challenging to solve. Furthermore, alternating optimization algorithms may fail to converge in such scenarios. Thirdly, the optimization results are only valid for the current environment. It may become ineffective and struggle to meet the requirements for real-time decision-making as the environment changes.

Though the research [11], [12], [13], [14], [15], [16], [17] applied the intelligent algorithms such as RL and DRL, the research of anti-jamming transmission for integrated air-ground networks still needs to be improved. Introducing RL or DRL technologies can endow autonomous distributed control of UAVs with stronger adaptability and flexibility, enhance the ability of individual UAVs to complete tasks, and also strengthen the coordination and cooperation of UAV swarms, thereby improving the overall performance of the system. In the UAV swarm system, the environmental information perceived by individual UAVs is local, so the strategy obtained through traditional single agent reinforcement learning algorithms is not universal. To solve this problem, multi-agent RL adds the number of agents based on single agent RL, and introduces a distributed collaborative strategy mechanism to make each agent autonomous, purposeful, and coordinated. Unlike traditional single agent RL, the state space, action space of multi-agents, and the number of agents is large and heterogeneous. Moreover, the environment is partially observable and unstable, and there are mutual influences and constraints among multiple agents and between agents and the environment. How to solve the aforementioned problems remains to be further studied and improved.

Existing literature [18], [19], [20], [21], [22] on the networking technology of UAV swarms in jamming environments is relatively limited. However, the strong adversarial nature and high dynamic characteristics in malicious jamming environments pose higher requirements for UAV networking technology. Therefore, studying autonomous intelligent networking strategies for UAV swarms is of great significance for enhancing the robustness and reliability of UAV communication networks.

C. Main Contributions

Inspired by the aforementioned motivations, this paper presents an anti-jamming transmission algorithm based on game theory and MADRL for the integrated air-ground network scenario, aiming to resist the both ground and air jammers and effectively enhance the regional coverage capability of the communication system. For uplink process, we propose UAV deployment schemes based on congestion game model and dynamic networking schemes based on coalition game, aiming to counteract malicious jamming from ground and air jammers, effectively enhancing the anti-jamming transmission

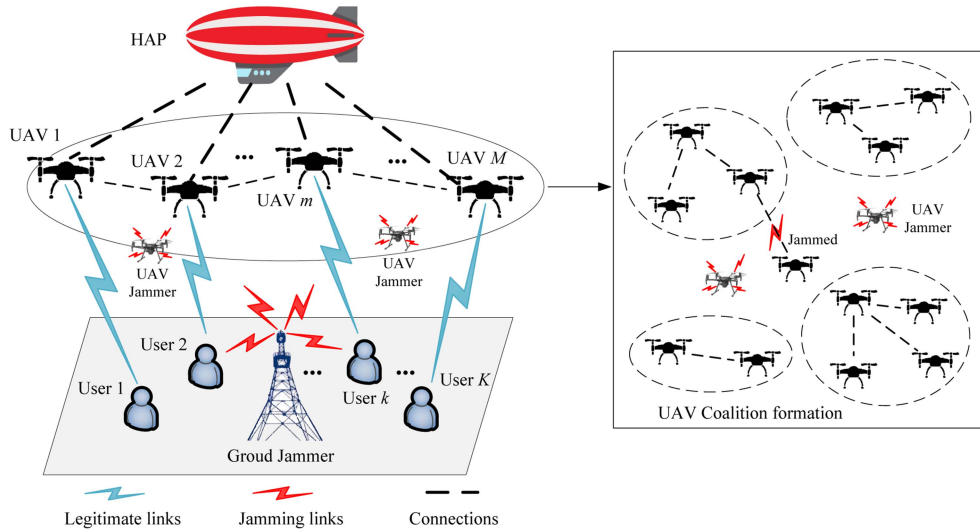


Fig. 1. System model.

capability of air-ground networks. For downlink process, the approach involves adopting a centralized training and distributed execution framework. To update the parameters of the soft actor-critic network, an improved proximal policy optimization (PPO) algorithm is proposed, which improves the effectiveness in complex multi-agent environments. The major contributions are summarized as follows.

- For the uplink process, we establish tasks for data collection and uploading by a swarm of UAVs. We construct a congestion game model for UAV swarm coverage, optimizing the deployment of UAV nodes to avoid ground jamming and increase data collection efficiency. In the dynamic networking phase, we build a coalition game model for UAV swarm formation, creating alliance subnets to avoid air jamming areas. Each UAV will transmit collected data to the cluster head UAV within the subnet, which will then upload the data to the HAP.
- For the downlink process, we design a joint power allocation and trajectory optimization problem for multi-UAV networks. Assuming that the jammer is intelligent, capable of perceiving the state of the UAV and dynamically optimizing the jamming strategy to maximize the utility function. The problem is transformed into a partially observable Markov decision process (POMDP) problem. Each UAV observes the users associated with itself, without needing to obtain complete information from the environment. We propose a MADRL-based anti-jamming transmission algorithm in multi-UAV collaboration scenarios, aiming to maximize the sum of all users' achievable sum rate with a lower power consumption.
- A centralized training and distributed execution framework is developed. During the centralized training stage, the experiences of each agent obtained from interacting with the environment are stored in an experience replay pool. During the distributed execution stage, each agent uses the trained soft actor-critic network to take actions and adjust its flight position and transmission power for joint service provision in a distributed manner.

- An improved PPO algorithm based on clip and count is proposed to update the soft actor-critic network parameters. Specifically, an improved fragment distribution algorithm is designed for the hard constraints between the new and old policies, which improves the effectiveness in complex multi-agent environments. Simulation results demonstrate that the proposed method outperforms traditional algorithms in terms of achieving higher system achievable sum rate, lower power consumption, and faster convergence speed.

We introduce the system model and the optimization problem for uplink and downlink process in Sections II and III. Analysis of the MADRL based anti-jamming transmission for multi-UAV network is presented in Section IV. Section V gives a framework of the MADRL algorithm with centralized training and distributed execution, along with the improved PPO algorithm and complexity analysis. Then, simulation results are presented in Section VI, and we conclude the whole paper in Section VII.

II. SYSTEM MODEL AND PROBLEM FORMULATION FOR UPLINK PROCESS

The system model is shown in Fig. 1. We consider a multi-UAV multi-user multi-input multi-output (MIMO) communication system with a HAP, M UAVs, K users, an intelligent ground jammer, and some air UAV jammers. Each UAV is equipped with N antennas, each air jammer and user is equipped with a single antenna, and the ground jammer is equipped with L antennas.¹ This paper mainly investigates the uplink transmission from

¹As for the unique advantages of HAPs compared to ground centralized servers, it can be concluded as follows. The deployment of HAP has strong flexibility and could provide a wider coverage. It is not limited by terrestrial conditions and has obvious advantages in connectivity for remote areas not served by terrestrial networks [23]. Moreover, HAP could carry servers and batteries, which can be used as a stable aerial base station to provide computing services and energy support for devices [24]. All of all, the HAP-assisted UAV communication system has the characteristics of low deployment cost, fast setup speed, high communication quality, strong adaptability, and wide coverage. It is not only used in our paper's situation but also widely used in military communication, disaster relief, emergency response.

users to UAVs under the air jamming condition, as well as the downlink transmission process from UAVs to users under the ground jamming condition. First, we introduce the issue of uplink transmission process.

For the uplink process, M UAVs operate in formation to conduct data collection tasks to the ground. Through coordinated deployment, data collection is performed, and the collected data is uploaded to the HAP by the UAV formation. In the UAV deployment stage and data collection stage, the UAV swarm optimizes the deployment of UAV nodes to avoid ground jammer and increase data collection efficiency. In the UAV dynamic networking stage and data upload stage, the UAV swarm forms alliance subnets to evade air jammers, and each UAV transmits collected data to the cluster head UAV within the subnet. Finally, in the data upload stage, the cluster head UAV aggregates the data and uploads it to the HAP.

A. UAV Deployment and Networking Problem Formulation

Discretize the ground area into X grids, and represent the grid position with the coordinates of the grid center. This chapter intends to first maximize the amount of data collection through collaborative location optimization $\{d_1(t), d_2(t), \dots, d_M(t)\}$ of UAV swarms. Assuming that the data volume follows a Poisson distribution, i.e. $\Gamma(g) \sim P(\lambda_g)$, the probability density distribution function is $\Pr[\Gamma(g) = k] = \frac{(\lambda_g)^k}{k!} e^{-\lambda_g}$. The actual amount of data collected by the UAV is $Q(u) = \sum_{i=1}^{|\Theta(u)|} \Gamma(g)/m(g)$, where $\Theta(u)$ represents a collection of grids collected by UAV. If the same grid is covered by $m(g)$ UAVs, due to mutual interference, the amount of data each UAV can collect is reduced to $\Gamma(g)/m(g)$. We aim to maximize the total amount of data collected by all UAVs through collaborative location optimization of the UAV swarm. The optimization problems can be modeled as

$$P1 : \max_{\{d_1(t), d_2(t), \dots, d_M(t)\}} \sum_{u=1}^M Q(u). \quad (1)$$

After the deployment of the UAV swarm, data is collected and transmitted to the HAP. Due to the presence of air jammers, the LoS transmission link between some UAVs and the HAP is disrupted under the threat of interference. In order to improve data upload performance, each UAV forms a dynamic subnet with neighboring UAVs, selects cluster head UAVs, and uniformly uploads data from the subnet to the HAP to reduce transmission loss and improve network robustness. Each UAV equipped with a positioning device can obtain the position information of adjacent UAV nodes, but can only communicate with nodes within a distance range. The set of adjacent UAV nodes that each UAV can communicate with is $A^i = \{u_i | \|d_i(t) - d_{i'}(t)\| \leq d_{th}\}, \forall u_i \in [1, M]$. The UAV cluster subnet Co_n consists of $\{u_0^n\} \cup \{u_i^n, i = 1, 2, \dots\}$, where u_0^n is the cluster head UAV, $\{u_i^n, i = 1, 2, \dots\}$ is the sub UAV. Define

$$\begin{aligned} |Co_n| &= \{u_0^n, u_i^n, \dots\} \\ s.t. \quad i &= 1, 2, \dots, |Co_n| - 1, \\ \|d_0^n - d_i^n\| &\leq d_{th}, \forall u_i^n \in Co_n, \end{aligned} \quad (2)$$

where $|Co_n|$ represents the potential of the set Co_n . The sub UAVs $\{u_i^n, i = 1, 2, \dots\}$ within the subnet Co_n upload

their respective data to the cluster head UAV u_0^n , which then uploads the aggregated data to the HAP. The Nakagami-m model [28] is used to model the UAV transmission channel $h_{0,i}$ and jamming channel $h_{j,0}^{J,n}$. The data upload rate between sub-UAVs $\{u_i^n, i = 1, 2, \dots\}$ and cluster head UAV u_0^n is $R_{0,i}^n = B \log_2(1 + \frac{P_m h_{0,i}}{\sum_{j=1}^{N_J} P_j^J h_{j,0}^{J,n} + \sigma_m^2})$, where B , P_m , P_j^J , σ_m^2 respectively represent the channel bandwidth, UAV transmission power, jamming power, and noise.

Since the channels of UAV-ground users may be obstructed by obstacles, we adopt the PLoS model in our paper. However, the channels of UAV-HAP are usually LoS links due to the high altitude. Considering the unique air-to-air communication characteristics between the UAV and HAP, we use the channel model according to [29] to describe the UAV-HAP communication. Combined with Shannon theory, the data transmission rate of the cluster head UAV u_0^n and the HAP is calculated by $R_0^n = B_{uh} \log_2(1 + \frac{p_0^n G_{uh} L_s L_l}{k_B T_s B_{uh}})$, where B_{uh} is the channel bandwidth between the cluster head UAV and the HAP, p_0^n is the transmission power of the cluster head UAV, G_{uh} is the antenna power gain, L_s is the total line loss, L_l is the free space loss, k_B is Boltzmann constant, and T_s is the system noise temperature.

We aim to choose the optimal subnet formation strategy to maximize the total amount of data uploaded by the UAV swarm to the HAP in time T in the interference environment. The optimization problem can be modeled as

$$\begin{aligned} P2 : \max_{\{Co_n\}_\pi} R_{total} \\ s.t. \quad R_{total} &= \sum_{n=1}^N \min\{(T - \Delta t) R_0^n, R_n\}, \\ R_n &= \sum_{i=1}^{|Co_n|} \min\{\Delta t \cdot R_{0,i}^n, C(u_i^n)\}, \end{aligned} \quad (3)$$

where N represents the total number of subnet alliances formed by the UAV group, Δt is the time spent in the UAV networking stage, R_n is the actual amount of data collected by the alliance $Co_n = \{u_0^n, u_i^n, \dots\}$ to the cluster head UAV u_0^n in time Δt , R_0^n is the data transmission rate between the cluster head UAV u_0^n and HAP, $\min\{(T - \Delta t) R_0^n, R_n\}$ is the total amount of data uploaded by the cluster head UAV u_0^n to the HAP, and R_{total} is the total amount of data up-loaded to all subnets formed by the UAV group.

B. Game Modeling and Equilibrium Analysis

This section models the optimization problem of UAV swarm node deployment in jamming environments as a congestion game model. The definition is as follows.

Definition 1 (Exact potential energy game): If there exists a potential energy function that satisfies $\phi(\pi_n, \pi_{-n}) - \phi(\pi_{n'}, \pi_{-n'}) = r(\pi_n, \pi_{-n}) - r(\pi_{n'}, \pi_{-n'})$, and when the UAV changes its strategy, the increment of its game utility is equal to the increment of the potential energy function, then the game model is an exact potential energy game. The exact potential game model has at least one pure strategy Nash equilibrium solution.

Theorem 1: The proposed UAV congestion game is an exact potential energy game, and there exists at least one stable Nash equilibrium solution. The proof is as follows.

Based on the Rosenthal function [25], the potential energy function is defined as $\Phi(u, -u) = \sum_{x \in [1, X]} \sum_{i=1}^{m(x)} \frac{\Gamma(x)}{i}$. When the position of the UAV changes, the change in utility function is

$$r_u - r_{u'} = \sum_{i \in \Theta(u)} \frac{\Gamma(i)}{m(i)} - \sum_{j \in \Theta(u')} \frac{\Gamma(j)}{m'(j)}. \quad (4)$$

There may be intersections between the sets $\Theta(u)$ and $\Theta(u')$. Let's assume the intersection is $\Xi = \Theta(u) \cap \Theta(u')$. In this case, for the grids in the intersection Ξ , the number of UAVs covering these grids remains unchanged, thus the amount of data that can be collected from these grids does not change, i.e., $\sum_{i \in \Xi} \Gamma(i)/m(i) = \sum_{i \in \Xi} \Gamma(i)/m'(i)$. So we have $r_u - r_{u'} = \sum_{i \in \{\Theta(u) \setminus \Xi\}} \frac{\Gamma(i)}{m(i)} - \sum_{j \in \{\Theta(u') \setminus \Xi\}} \frac{\Gamma(j)}{m'(j)}$, where $\{\Theta(u) \setminus \Xi\}$ represents the collection of grids previously covered but no longer covered. For $k \in \{\Theta(u) \setminus \Xi\}$, it can be derived that $m'(k) = m(k) - 1$. Similarly, for $k \in \{\Theta(u') \setminus \Xi\}$, it can be derived that $m'(k) = m(k) + 1$. When the UAV's position changes, the potential function changes to

$$r_u - r_{u'} = \sum_{i \in \{\Theta(u) \setminus \Xi\}} \frac{\Gamma(i)}{m(i)} - \sum_{j \in \{\Theta(u') \setminus \Xi\}} \frac{\Gamma(j)}{m'(j)}. \quad (5)$$

Hence, the change in the potential function is as (6) shown at the bottom of this page.

Since $r_u - r_{u'} = \Phi(u, -u) - \Phi(u', -u')$, according to the definition of exact potential [25], the proposed game model is an exact potential game, so there is at least one stable Nash equilibrium solution.

Based on the above analysis, the deployment algorithm for UAVs based on congestion game is shown in Algorithm 1 to solve P1.

After the deployment of UAV nodes is completed, the dynamic networking problem of UAVs is modeled as an alliance game. The definition is as follows.

Definition 2 (Alliance game): The UAV swarm $\mathcal{M} \in \{1, 2, \dots, M\}$ can be divided into multiple non overlapping alliances $\{C_{o_n}\}$ that satisfy $\bigcup_{n=1}^N C_{o_n} = \mathcal{M}$. Each UAV has its own alliance preference criterion $\succ = (\succ_1, \succ_2, \dots, \succ_N)$, which $\succ_i$ represents the preference ranking of UAVs u_i choosing to join the alliance, and the strategy constitutes an alliance game.

The UAV swarm forms a dynamic subnet through alliance game, and the UAVs in the subnet upload the collected data to the cluster head UAV, which then uploads it to

Algorithm 1: Congestion Game Based UAV Deployment Algorithm.

Input: The initial position of the UAVs, the position of ground jammer, the grid target area, and the distribution of data volume in each grid.

Output: Optimized deployment location of UAVs.

- 1: **for** each time slot $t = 1, \dots, T$ **do**
 - 2: **for** each UAV $m = 1, \dots, M$ **do**
 - 3: Calculate the total amount of data that can be collected by all the current UAVs.
 - 4: Randomly select a UAV, choose a direction of motion (up, down, left, right, left front, left rear, right front, right rear), and update the amount of data that can be collected for each grid.
 - 5: If the total amount of data that can be collected increases, update the location; otherwise, maintain the status situation.
 - 6: **end for**
 - 7: **end for**
-

the nearby launch platform. Define the total amount of data uploaded by each UAV in the alliance C_{o_n} as $E(C_{o_n}) = \max\{\min\{(T - \Delta t)R_0^n, \mathcal{R}_n\}\}$. Where R_0^n represents the upload rate of the cluster head UAV that can achieve the maximum amount of uploaded data, and $\mathcal{R}_n$ is the total amount of data aggregated to the cluster head UAV in C_{o_n} . The alliance utility function for UAV u_i to select C_{o_n} is defined as $\varphi_i = E(C_{o_n} \cup u_i) - E(C_{o_n})$. Where $E(C_{o_n} \cup u_i)$ represents the total amount of data uploaded by the alliance after the UAV u_i joins C_{o_n} . The alliance utility function can characterize the improvement effect of UAV u_i joins the alliance C_{o_n} on the total amount of data uploads.

Definition 3 (Win win criterion): Each UAV node selects its alliance according to the win-win criterion. If the alliance selection of the node $u_i \in C_{o_{n'}}$ changes from $C_{o_{n'}}$ to C_{o_n} , the sum utility of other UAVs $C_{o_{n'}/u_i}$ in the original alliance and UAVs $C_{o_n} \cup u_i$ in the new alliance increases, and this change is established, that is

$$C_{o_n} \succ_i C_{o_{n'}} \Leftrightarrow E(C_{o_n} \cup u_i) + E(C_{o_{n'} \setminus u_i}) > E(C_{o_n}) + E(C_{o_{n'}}), \quad (7)$$

$$\begin{aligned} \Phi(u, -u) - \Phi(u', -u') &= \sum_{x \in \{\Theta(u) \setminus \Xi\} \cup \{\Theta(u') \setminus \Xi\}} \sum_{i=1}^{m(x)} \frac{\Gamma(x)}{i} - \sum_{y \in \{\Theta(u) \setminus \Xi\} \cup \{\Theta(u') \setminus \Xi\}} \sum_{j=1}^{m'(y)} \frac{\Gamma(y)}{j} \\ &= \sum_{x \in \{\Theta(u) \setminus \Xi\}} \sum_{i=1}^{m(x)} \frac{\Gamma(x)}{i} + \sum_{x \in \{\Theta(u') \setminus \Xi\}} \sum_{i=1}^{m(x)} \frac{\Gamma(x)}{i} - \sum_{y \in \{\Theta(u) \setminus \Xi\}} \sum_{j=1}^{m'(y)} \frac{\Gamma(y)}{j} - \sum_{y \in \{\Theta(u') \setminus \Xi\}} \sum_{j=1}^{m'(y)} \frac{\Gamma(y)}{j} \\ &= \sum_{x \in \{\Theta(u) \setminus \Xi\}} \left(\sum_{i=1}^{m(x)} \frac{\Gamma(x)}{i} - \sum_{j=1}^{m'(x)} \frac{\Gamma(x)}{j} \right) + \sum_{y \in \{\Theta(u') \setminus \Xi\}} \left(\sum_{i=1}^{m(y)} \frac{\Gamma(y)}{i} - \sum_{j=1}^{m'(y)} \frac{\Gamma(y)}{j} \right) \\ &= \sum_{x \in \{\Theta(u) \setminus \Xi\}} \left(\sum_{i=1}^{m(x)} \frac{\Gamma(x)}{i} - \sum_{i=1}^{m(x)-1} \frac{\Gamma(x)}{i} \right) + \sum_{y \in \{\Theta(u') \setminus \Xi\}} \left(\sum_{j=1}^{m(y)} \frac{\Gamma(y)}{j} - \sum_{j=1}^{m(y)+1} \frac{\Gamma(y)}{j} \right) \\ &= \sum_{x \in \{\Theta(u) \setminus \Xi\}} \frac{\Gamma(x)}{m(x)} - \sum_{y \in \{\Theta(u') \setminus \Xi\}} \frac{\Gamma(y)}{m(y)+1} = \sum_{x \in \{\Theta(u) \setminus \Xi\}} \frac{\Gamma(x)}{m(x)} - \sum_{y \in \{\Theta(u') \setminus \Xi\}} \frac{\Gamma(y)}{m'(y)}. \end{aligned} \quad (6)$$

where $E(Co_n)$ represents the sum speed of all UAVs in the alliance Co_n . According to the win-win principle, if a UAV $u_i \in Co_{n'}$ joins a new alliance Co_n from the old alliance $Co_{n'}$, the total amount of data uploaded by all UAVs in the new and old alliances will increase.

Definition 4 (Stable alliance formation): If there is no UAV that can unilaterally improve its own game effectiveness by changing its alliance formation strategy, then the current alliance formation plan is stable, that is

$$\varphi_i(Co_n, Co_{-n}) \geq \varphi_i(Co_{n'}, Co_{-n'}), \forall u_i \in \mathcal{M}, Co_n \neq Co_{n'}. \quad (8)$$

Next, with the help of precise potential game theory, it is proven that the stable alliance formation strategy obtained is a Nash equilibrium solution. Define the potential energy function as the total amount of uploaded data for all alliances, i.e., $\Psi = \sum_{n=1}^N E(Co_n)$. When the UAV u_i joins the new alliance Co_n from the old alliance $Co_{n'}$, the change of utility function is

$$\begin{aligned} & \varphi_i(Co_n) - \varphi_i(Co_{n'}) \\ &= [E(Co_n \cup u_i) - E(Co_n)] - [E(Co_{n'}) - E(Co_{n'} \setminus u_i)] \end{aligned} \quad (9)$$

Due to the fact that the alliance change of UAV u_i only affect the old and new alliances, and have no impact on other alliances, thus $\sum_{k \in [1, N] \setminus n \setminus n'} E(Co_k) = \sum_{k' \in [1, N] \setminus n \setminus n'} E(Co_{k'})$. The change of utility function is

$$\begin{aligned} & \Psi_i(Co_n) - \Psi_i(Co_{n'}) \\ &= E(Co_n \cup u_i) + E(Co_{n'} \setminus u_i) - E(Co_n) - E(Co_{n'}) \\ &= [E(Co_n \cup u_i) - E(Co_n)] - [E(Co_{n'}) - E(Co_{n'} \setminus u_i)] \end{aligned} \quad (10)$$

Comparing formulas (9) and (10), it can be concluded that $\Psi_i(Co_n) - \Psi_i(Co_{n'}) = \varphi_i(Co_n) - \varphi_i(Co_{n'})$. Hence the proposed alliance game is an exact potential energy game with at least one stable Nash equilibrium solution.

To solve P2, we propose a UAV networking algorithm based on alliance game, where each UAV selects alliances and dynamically forms multiple stable subnets based on the alliance utility and win-win criteria. The specific algorithm process is shown in Algorithm 2.

III. SYSTEM MODEL AND PROBLEM FORMULATION FOR DOWNLINK PROCESS

After the formation of multiple stable subnets by the UAV swarm, let us proceed to discuss the issue of downlink transmission. For the downlink transmission, UAVs transmit legitimate signals to users while the ground jammer release interference signals to users within a given power range, which reduces the SINR received by users, affecting the quality of communication. The HAP is used to train the UAV network, reducing the internal burden of UAVs. The aim is to change the UAVs' flight trajectory to keep them as far away as possible from the jammer and fly around the users and dynamically adjust their

Algorithm 2: Alliance Game Based UAV Networking Algorithm.

Input: The current location of the UAV swarm, the current data collection volume, the location of airborne jammers, and the data upload capabilities of each UAV.
Output: The networking results of UAV clusters and cluster head UAVs after the alliance formation.

- 1: Initialize alliance groups, each containing only one UAV.
- 2: **for** each time slot $t = 1, 2, \dots, \Delta t$ **do**
- 3: Calculate the total amount of data that can be collected by all current UAVs. Randomly select a UAV, select the cluster head UAV in its alliance, transmit the collected data to the cluster head UAV, upload the summary data to the cluster head UAV, and calculate the alliance utility of the UAV.
- 4: The UAV chooses to join neighboring alliances and calculates the new alliance utility.
- 5: According to the win-win principle, if the UAV can achieve higher alliance utility, the alliance will be changed. Otherwise, the current alliance will remain unchanged.
- 6: **end for**

transmission power within a period of time T to maximize the users' achievable sum rate.²

A. System Model

1) **Channel Model:** We adopt a more practical probabilistic line of sight (PLoS) model based on UAV flight altitude and elevation angle [26]. Assuming that the UAVs fly at a uniform speed with a fixed height in three-dimensional space, and the positions of the user and the jammer remain fixed. At time t , the three-dimensional position coordinates of UAV m , user k , and jammer are represented as $d_m(t) = [x_m(t), y_m(t), h_m]$, $d_k = [x_k, y_k, 0]$, and $d_J = [x_J, y_J, 0]$, respectively, where h_m represents the flight height of the UAV m . Using the PLoS channel model, the probability of a LoS channel between the UAV m and user k is

$$P_{m,k}^{LoS}(t) = \frac{1}{1 + a \exp(-b(\frac{180}{\pi} \sin^{-1}(\frac{h_m}{d_{m,k}(t)}) - a))}, \quad (11)$$

where a and b are constants depending on the environment, $d_{m,k}(t) = \sqrt{(x_m(t) - x_k)^2 + (y_m(t) - y_k)^2 + h_m^2}$ represents the distance between UAV m and user k at time t . Therefore, the probability of a non-line of sight (NLoS) channel between UAV m and user k is $P_{m,k}^{NLoS}(t) = 1 - P_{m,k}^{LoS}(t)$. The path loss for LoS and NLoS channels are

²To accurately reflect realistic operational scenarios, we consider jammers as both ground stationary and air dynamic nodes. The ground base station acts as a stationary jammer, interfering with the downlink signal reception of ground users. The mobile UAV serves as a dynamic jammer, interfering with the uplink network formation of aerial nodes. The jammers are with sufficient power resources to provide a more realistic and challenging testbed for the proposed algorithm.

represented as $L_{m,k}^{LoS}(t) = L_{m,k}^{FS}(t) + \eta_{m,k}^{LoS}$ and $L_{m,k}^{NLoS}(t) = L_{m,k}^{FS}(t) + \eta_{m,k}^{NLoS}$, where $L_{m,k}^{FS}(t) = 20 \log(4\pi f d_{m,k}(t)/c)$ is the free space path loss from UAV m to user k . In addition, $\eta_{m,k}^{LoS}$ and $\eta_{m,k}^{NLoS}$ represent the average additional path loss. Therefore, the average path loss between UAV m and user k at time t can be represented as

$$\beta_{m,k}(t) = P_{m,k}^{LoS}(t)L_{m,k}^{LoS}(t) + P_{m,k}^{NLoS}(t)L_{m,k}^{NLoS}(t). \quad (12)$$

Therefore, the channel model $\mathbf{h}_{m,k}(t) \in \mathbb{C}^{N \times 1}$ between UAV m and user k can be represented as [27]

$$\mathbf{h}_{m,k}(t) = \beta_{m,k}^{1/2}(t)\mathbf{g}_{m,k}(t), \quad (13)$$

where $\mathbf{g}_{m,k}(t) = [g_{m,k,1}(t), \dots, g_{m,k,n}(t), \dots, g_{m,k,N}(t)]^T \in \mathbb{C}^{N \times 1}$ is the small-scale fading vector. $g_{m,k,n}(t)$ is the small-scale fading coefficient between the antenna n of UAV m and user k , which can be represented as

$$g_{m,k,n}(t) = \sqrt{\frac{K_a}{K_a + 1}} e^{-j\frac{2\pi}{\lambda} d_{m,k}(t)} + \sqrt{\frac{1}{K_a + 1}} p_{m,k,n}(t), \quad (14)$$

where K_a is the Rician factor, λ is the carrier wavelength, $p_{m,k,n}(t) \sim \mathcal{CN}(0, 1)$ is Rayleigh fading. Therefore, the channel $\mathbf{H}_m(t) \in \mathbb{C}^{K \times N}$ of the UAV m to the all users ($k = 1, 2, \dots, K$) is given by

$$\mathbf{H}_m(t) = \mathbf{B}_m^{1/2}(t)\mathbf{G}_m(t), \quad (15)$$

where $\mathbf{B}_m(t) \in \mathbb{C}^{K \times K}$ is the large-scale fading matrix, and the k -th element on the diagonal matrix is $[\mathbf{B}_m(t)]_{k,k} = \beta_{m,k}(t)$. $\mathbf{G}_m(t) = [\mathbf{g}_{m,1}(t), \dots, \mathbf{g}_{m,k}(t), \dots, \mathbf{g}_{m,K}(t)]^T \in \mathbb{C}^{K \times N}$ is the fast fading matrix.

The channel $\mathbf{h}_{J,k} \in \mathbb{C}^{L \times 1}$ between the jammer and user k is given by $\mathbf{h}_{J,k} = \sqrt{\beta_{J,k}}\mathbf{g}_{J,k}$, where $\beta_{J,k} = \psi\tau_0/d_{J,k}^\kappa$ represents the large-scale fading between the jammer and user k and $\psi \sim \mathcal{LN}(0, \sigma_s)$ represents shadow fading. τ_0 is the path loss at the reference distance. $d_{J,k}$ denotes distance between the jammer and user k , which can be expressed as $d_{J,k} = \sqrt{(x_J - x_k)^2 + (y_J - y_k)^2}$. κ is the path loss exponent. $\mathbf{g}_{J,k} \in \mathbb{C}^{L \times 1} \sim \mathcal{CN}(0, I_L)$ represents Rayleigh fading. Therefore, the channel $\mathbf{H}_J \in \mathbb{C}^{K \times L}$ of the jammer to the all users $k = 1, 2, \dots, K$ can be represented as $\mathbf{H}_J = \mathbf{B}_J^{1/2}\mathbf{G}_J$, where $\mathbf{B}_J \in \mathbb{C}^{K \times K}$ is a diagonal matrix and $\mathbf{G}_J = [\mathbf{g}_{J,1}, \dots, \mathbf{g}_{J,k}, \dots, \mathbf{g}_{J,K}]^T \in \mathbb{C}^{K \times L}$ is the fast fading matrix.

2) *Signal Transmission Model*: For downlink process, we optimize the power allocation and trajectory of the UAV within a period of time T , and at any time t , zero force precoding³ and user equal power allocation are adopted to reduce computational complexity [27], [30]. Define $\mathbf{W}_m(t) \in \mathbb{C}^{N \times K}$ and $\mathbf{W}_J \in \mathbb{C}^{L \times K}$ as the precoding matrix for the UAV m and jammer, which is given by

$$\mathbf{W}_m(t) = \frac{\sqrt{1/\|\mathbf{H}_m(t)\mathbf{H}_m^H(t)\|^{-1}}\mathbf{H}_m^H(t)}{\mathbf{H}_m(t)\mathbf{H}_m^H(t)}, \quad (16)$$

³Note that this paper aims to utilize MADRL technology to address trajectory optimization and power allocation issues, with precoding not being the focus of this study. Existing literature, such as [31], [32], [33], [34], has conducted in-depth research on obtaining optimal precoding for UAV-assisted system. Therefore, this paper adopts zero-forcing precoding for UAVs and jammers to reduce computational complexity.

and

$$\mathbf{W}_J = \frac{\sqrt{1/\|(\mathbf{H}_J\mathbf{H}_J^H)^{-1}\|^2}\mathbf{H}_J^H}{\mathbf{H}_J\mathbf{H}_J^H}, \quad (17)$$

respectively. Then the received signal at user k can be expressed as

$$y_k(t) = \sum_{m=1}^M \sqrt{P_m(t)/K} \mathbf{h}_{m,k}^T(t) \mathbf{W}_m(t) \mathbf{x}_m + \sqrt{P_J(t)/K} \mathbf{h}_{J,k}^T \mathbf{W}_J \mathbf{x}_J + z_k, \quad (18)$$

The first term represents the signals of all UAVs ($m = 1, 2, \dots, M$), which need to be included in the summation. The second term is the signal of the jammer, and the third term is noise, which do not need to be included in the summation. $\mathbf{x}_m = [x_{m,1}, \dots, x_{m,K}]^T \in \mathbb{C}^{K \times 1}$ is the signal vector transmitted by UAV m to all users, $\mathbf{x}_J = [x_{J,1}, \dots, x_{J,K}]^T \in \mathbb{C}^{K \times 1}$ is the jamming signal transmitted by the jammer, $P_m(t)$ represents the transmit power of UAV m , $P_J(t)$ represents the interference power of the jammer, and $z_k \sim \mathcal{CN}(0, \sigma^2)$ represents the Gaussian white noise. Then, the received SINR at user k from UAV m at time t can be expressed as [30]

$$\gamma_{m,k}(t) = \frac{P_m(t)/K \left[(\mathbf{H}_m(t)\mathbf{H}_m^H(t))^{-1} \right]_{k,k}}{P_J(t)/K \left[(\mathbf{H}_J\mathbf{H}_J^H)^{-1} \right]_{k,k} + \sigma_{m,k}^2}. \quad (19)$$

The achievable rate obtained by user k from all the UAVs ($m = 1, 2, \dots, M$) at time t can be expressed as

$$R_k(t) = \sum_{m=1}^M B \log_2 [1 + \gamma_{m,k}(t)]. \quad (20)$$

For reference purposes, the main notions and descriptions are given in Table I.

Note that the channels mentioned above, $\mathbf{H}_m(t)$ and $\mathbf{H}_J$, are required to obtain the full channel knowledge necessary for computing the precoding matrix and received signal. In cases where the ideal assumption does not hold, we can obtain the CSI among the UAV, jammer and users by utilizing the existing channel estimation method. However, there inevitably exist channel estimation errors between the actual channel matrix and the imperfect or estimated channel matrix. These errors can be modeled as circularly symmetric complex Gaussian (CSCG) variables [35], [36], [37]. Taking the jammer channel $\mathbf{H}_J$ as an example, when considering channel estimation errors, the actual CSI can be expressed as $\mathbf{H}_J = \hat{\mathbf{H}}_J + \Delta\mathbf{H}_J$. $\mathbf{H}_J$, $\hat{\mathbf{H}}_J$, and $\Delta\mathbf{H}_J$ are the actual channel, estimated channel, and channel estimation error, respectively. $\Delta\mathbf{H}_J$ can be modeled as CSCG variables, i.e., $\Delta\mathbf{H}_J \sim \mathcal{CN}(0, \mathbf{A}_J \otimes \mathbf{B}_J)$. $\mathbf{A}_J$ and $\mathbf{B}_J$ are the channel estimation error covariance matrices of the user and jammer, respectively [35], [36]. Assuming that channel estimation errors are independent of the estimated channel matrices and their statistics are known to the transceivers, we thus have

$$\mathbf{H}_J \sim \mathcal{CN}(\hat{\mathbf{H}}_J, \mathbf{A}_J \otimes \mathbf{B}_J). \quad (21)$$

Similarly, $\mathbf{H}_m(t)$ could be obtained using the above method. By substituting the actual $\mathbf{H}_m(t)$ and $\mathbf{H}_J$ into (19), the achievable rate under imperfect CSI can be obtained.

TABLE I
NOTIONS AND DESCRIPTIONS

Notions	Descriptions
h_m	UAV flying height
$v_m(t)$	UAV flight speed
$d_{\min}$	UAV minimum distance
$P_m(t)$	UAV power
$P_J(t)$	Ground jammer power
τ_0	Path loss
κ	Path loss exponent
δ	Learning rate
$d_m(t)$	Position coordinates of UAV m
d_J	Position coordinates of jammer
$d_{m,k}(t)$	Distance between UAV m and user k
$L_{m,k}^{\text{LoS}}(t)$	Path loss for LoS channels
$L_{m,k}^{\text{NLoS}}(t)$	Path loss for NLoS channels
$P_{m,k}^{\text{LoS}}(t)$	Probability of LoS channel between UAV m and user k
$P_{m,k}^{\text{NLoS}}(t)$	Probability of NLoS channel between UAV m and user k
$\mathbf{h}_{m,k}(t) \in \mathbb{C}^{N \times 1}$	Channel between UAV m and user k
$\mathbf{g}_{m,k}(t) \in \mathbb{C}^{N \times 1}$	Small-scale fading vector between UAV m and user k
$\mathbf{H}_m(t) \in \mathbb{C}^{N \times N}$	Channel of UAV m to the all users
$\mathbf{B}_m(t) \in \mathbb{C}^{K \times K}$	Large-scale fading matrix of UAV m
$\mathbf{G}_m(t) \in \mathbb{C}^{K \times N}$	Fast fading matrix of UAV m
$\mathbf{h}_{J,k} \in \mathbb{C}^{L \times 1}$	Channel between jammer and user k
$\mathbf{g}_{J,k} \in \mathbb{C}^{L \times 1}$	Small-scale fading vector between jammer and user k
$\mathbf{H}_J \in \mathbb{C}^{K \times L}$	Channel of UAV m to the all users
$\mathbf{B}_J \in \mathbb{C}^{K \times K}$	Large-scale fading matrix of the jammer
$\mathbf{G}_J \in \mathbb{C}^{K \times L}$	Fast fading matrix of the jammer
$\mathbf{W}_m(t) \in \mathbb{C}^{N \times K}$	Precoding matrix of the UAV m
$\mathbf{W}_J \in \mathbb{C}^{L \times K}$	Precoding matrix of the jammer
$y_k(t)$	Received signal of user k
$\gamma_{m,k}(t)$	Received SINR of the user k from UAV m
$R_k(t)$	Achievable rate of user k from all the UAVs

B. UAV Optimization Problem Formulation

The research objective is to optimize the UAV trajectory and dynamically adjust the transmission power to maximize the achievable sum rate $\sum_{k=1}^K R_k(t)$ of all users in the system. Based on the above assumptions, the following optimization problem is constructed.

$$\begin{aligned}
 & \max_{\mathcal{P}, \mathcal{D}} \sum_{k=1}^K R_k(t), \\
 & \text{s.t. } 0 \leq t \leq T, \\
 & \|d_m(t) - d_{m'}(t)\|_2 \geq d_{\min}, m \neq m', m = 1, \dots, M, \\
 & v_{\min} \leq v_m(t) \leq v_{\max}, m = 1, \dots, M, \\
 & d_m(1) = d_m(T), m = 1, \dots, M, \\
 & P_{\min}^M \leq P_m(t) \leq P_{\max}^M, m = 1, \dots, M, \\
 & P_{\min}^J \leq P_J(t) \leq P_{\max}^J,
 \end{aligned} \tag{22}$$

where $\mathcal{P} = P_m(t), m = 1, \dots, M$ represents the power set of the UAV. $\mathcal{D} = d_m(t), m = 1, \dots, M$ represents the trajectory set of the UAV. $0 \leq t \leq T$ indicates that this optimization problem is within a time period. $\|d_m(t) - d_{m'}(t)\|_2 \geq d_{\min}$, and $d_{\min}$ is the minimum distance limitation to ensure flight safety. $v_{\min} \leq v_m(t) \leq v_{\max}$ is the flight speed limitation of each UAV. $d_m(1) = d_m(T)$ is to ensure that after time T the UAV will return to the initial position. $P_{\min}^M \leq P_m(t) \leq P_{\max}^M$ and $P_{\min}^J \leq P_J(t) \leq P_{\max}^J$ indicate that the power of the UAVs and the jammer are within a certain range, where $P_{\min}^M$ and $P_{\max}^M$ represents the minimum and maximum selectable power for the UAVs, $P_{\min}^J$ and $P_{\max}^J$ represents the minimum and maximum selectable power for the jammer, respectively.

C. Jamming Problem Formulation

Assuming that the ground jammer has certain learning and reasoning capabilities. It can adjust its jamming strategy based

on real-time communication conditions and transmission power to achieve better interference effects. The paper models the jamming problem as a Markov decision process (MDP), represented by $\{S_J, A_J, R_J\}$, where S_J, A_J, R_J denote the state space, action space, and reward space, respectively. The intelligent jammer first observes the transmission power of the UAV and selects an appropriate jamming power $P_J(t) \in [P_{\min}^J, P_{\max}^J]$ using the Q -learning method. The specific model is as follows.

State: State serves as the basis for the action taken by the agent. In this paper, the jammer dynamically adjusts its jamming power based on the observation of the UAV's transmission power. Therefore, the state of the jammer at time t is determined based on the estimated transmit power of the UAV at time $t - 1$, i.e., $S_J(t) = \hat{P}_m(t - 1), m = 1, \dots, M$.

Action: This paper mainly considers power domain anti-jamming, that is, the only adjustable parameter for the jammer is jamming power. Therefore, the jammer's action is set to the jamming power $P_J(t)$, i.e., $A_J(t) = P_J(t)$.

Reward function: The primary goal of the jammer is to reduce the SINR of the users and disrupt the normal communication of legitimate links. At the same time, it aims to minimize the jammer's transmission power, reduce energy consumption, and improve invisibility. Taking all these factors into consideration, the reward function can be shown as follows

$$R_J(t) = -w_1 \sum_{k=1}^K R_k(t) - w_2 P_J(t), \tag{23}$$

where w_1 and w_2 are weight parameters that represent the relative importance of the user's received SINR and jamming power in the reward function. By designing the reward function, the jammer will select an appropriate jamming power based on the current state to maximize the reward function and achieve jamming objectives. Additionally, since the reward function takes into account the impact of transmission power, the jammer will also strive to minimize the transmission power to save energy and improve jamming invisibility.

IV. MADRL BASED ANTI-JAMMING TRANSMISSION FOR MULTI-UAV NETWORK

The game theory and MADRL-based anti-jamming transmission framework is given in Fig. 2. For the uplink process, we propose UAV deployment schemes based on congestion game model and dynamic networking schemes based on coalition game, aiming to counteract malicious jamming from ground and air jammers, effectively enhancing the anti-jamming transmission capability of air-ground networks. For the downlink process, we introduce the DRL method, treating each UAV as an agent and transforming the optimization problem into a POMDP. In essence, we firstly use KDE method to estimate the jammer transmit power. Using this estimate, the POMDP is formulated for UAV trajectory and transmit power optimization, with reward function incorporating the sum rate of the user, the distance of UAVs to the users and jammer, as well as the penalty factors for the UAV flying range and transmit power limitation. In order to handle the large state space, a centralized training and distributed execution approach is proposed, whereby HAP is used to run the centralized training from the experience replay

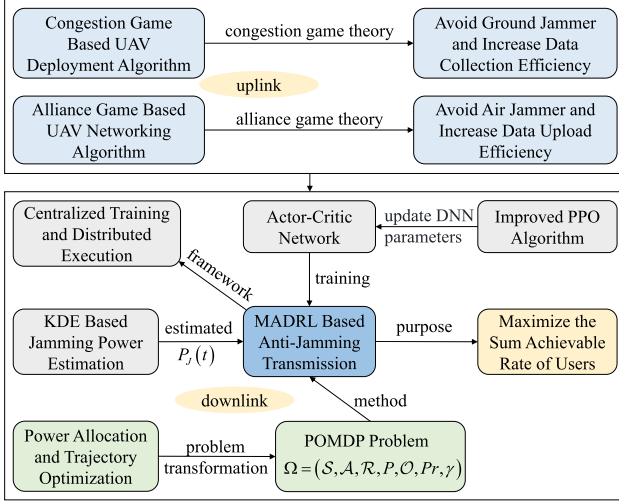


Fig. 2. Game theory and MADRL-based anti-jamming transmission framework.

buffer created at each of the UAVs, and the UAVs run the action sequence according to the actor network in a distributed fashion. Improved PPO is then used to allow the clipping parameter to be adapted to accelerate the algorithm convergence.

A. Effective Jamming Signal Strength Estimation

From formulas (19) and (20), it can be seen that to solve the optimization problem (22), it is first necessary to obtain the jamming power $P_J(t)$. In practical communication systems, estimating the jamming signal strength is crucial for optimizing the anti-jamming strategies of UAVs. However, due to the time-varying nature of wireless channels, the jamming signal strength may fluctuate and contain outliers, which will affect the training effectiveness of anti-jamming strategy selection. To address this issue, this section proposes a method based on KDE [27]. By fitting the distribution pattern of historical data of jamming signal strength and handling outliers, the variance of effective jamming signal strength can be reduced, thereby providing better guidance for UAV strategy updates.

Specifically, define the jamming signal strength of user k at time $t-1$ as $P_{J,k}(t-1) = \frac{P_{J,k}(t-1)}{K[(\mathbf{H}_J \mathbf{H}_J^H)^{-1}]_{k,k}}$. The jamming signal strength $P_{J,k}(t)$ is related to the actual jamming power and the jamming channel. The user can estimate the received SINR and provide feedback to the transmitter to obtain information about the jamming signal strength. The sum of jamming signal strengths for all users in the region is defined as $\bar{P}_J(t-1) = \sum_{k=1}^K P_{J,k}(t-1)$. Firstly, the historical data of jamming signal strengths is compressed using a logarithmic function

$$\tilde{P}_J(j) = 10 \lg \bar{P}_J(j), 1 \leq j \leq t-1. \quad (24)$$

Then the probability density function (PDF) of the logarithmically compressed historical data of jamming signal strengths can be obtained by KDE technique. This helps identify and handle outliers. Through the aforementioned steps, jamming signal strength data can be effectively processed in situations where wireless channels experience rapid changes, thereby better coping with the variations caused by the movement of the jammer.

Algorithm 3: Effective Jamming Power Estimation Based on KDE.

Input: Jamming signal strength level N_{num} , kernel function $G(x)$, parameter η , actual jamming signal strength $\tilde{P}_J(t-1)$ measured at time $t-1$.

Output: Effective jamming signal strength $\hat{P}_J(t)$.

- 1: Logarithmically compress the historical data of jamming signal strength $\tilde{P}_J(j)$ according to formula (24).
- 2: Obtain the estimated PDF of $\tilde{P}_J(j)$ according to formula (25).
- 3: Obtain the main value interval $[\tilde{P}_{\min}^J, \tilde{P}_{\max}^J]$ according to formula (26).
- 4: Discretize the main value interval with a quantization interval of ξ .
- 5: Determine the estimated effective jamming power $\hat{P}_J(t)$ of the next time according to formula (27).

KDE is a commonly used non-parametric method for probability density estimation [38], [39], which can estimate unknown PDF by smoothing the data. Assuming a set of independent and identically distributed random variable samples $x_1, x_2, \dots, x_n$ from an unknown PDF $f(x)$, the following formula can be used to estimate the PDF:

$$\hat{f}(x) = \frac{1}{nB_f} \sum_{i=1}^n G\left(\frac{x-x_i}{B_f}\right), \quad (25)$$

where $G(\cdot)$ is the kernel function, commonly chosen as the Gaussian kernel with the expression $G(x) = 1/\sqrt{2\pi}e^{-\frac{1}{2}x^2}$ [40]. B_f is the bandwidth parameter of the kernel function, which controls the smoothing level of the estimated PDF.

After obtaining the estimated PDF based on KDE, the main value range of the compressed jamming strength data is defined as $[\tilde{P}_{\min}^J, \tilde{P}_{\max}^J]$, which satisfies the following conditions:

$$\int_{\tilde{P}_{\min}^J}^{\tilde{P}_{\max}^J} \hat{f}_h(x) dx \geq \eta \int_{-\infty}^{+\infty} \hat{f}_h(x) dx, \quad (26)$$

where η is a constant close to 1. Based on the estimated PDF, the UAV discretizes the main value range with a quantization interval of $\xi = (\tilde{P}_{\max}^J - \tilde{P}_{\min}^J)/(N_{num} - 1)$. Finally, the effective interference signal strength at time t is obtained as

$$\hat{P}_J(t) = (\tilde{P}_J(t) - \tilde{P}_{\min}^J)/\xi, \quad (27)$$

Finally, based on the estimated PDF, the UAV can discretize the main value range to obtain the estimated effective jamming power.⁴ The specific process is shown in Algorithm 3.

B. MADRL Algorithm Based on POMDP

After obtaining the jamming power based on Algorithm 3, the next step is to solve the optimization problem (22). The objective function is non-convex, and there exists a coupling relationship between the achievable rate of the user and the position of the UAV, making (22) a challenging optimization problem that cannot be easily solved using traditional convex

⁴To run the KDE method, each user needs to report the jamming signal strength. The user can estimate the received SINR and provide feedback to the transmitter about the jamming signal strength. In this case, the noise power is assumed to be negligible compared to the jamming power.

optimization techniques. The POMDP is used to model random decision problems with incomplete observations [42]. Each UAV is considered as an agent, learning from and updating experiences based on the environment. Since each UAV can only observe information relevant to itself, the optimization problem (22) can be reformulated into a POMDP problem, specifically represented as

$$\Omega = (\mathcal{S}, \mathcal{A}, \mathcal{R}, P, \mathcal{O}, Pr, \gamma), \quad (28)$$

where $\mathcal{S}$, $\mathcal{A}$, $\mathcal{R}$, and $\mathcal{O}$ denote the state space, action space, reward space, and observation space of the UAV, respectively. P is the probability transition function for state s_t transition to the next state s_{t+1} . In a POMDP, the UAV receives observation values o_t from the observation space $\mathcal{O}$ rather than the state space $\mathcal{S}$. Pr is the observable conditional probability, which captures the uncertainty of the observation process. $\gamma \in [0, 1]$ is the reward discount factor. The detailed definitions of the modeled POMDP are given as follows.

State and observation: Each agent decides its actions based on the observed state information of the environment. The state space can be represented as

$$s_t = \{R_k(t), d_m(t), v_m(t), P_J(t)\}, \quad (29)$$

where $R_k(t)$ is the user achievable rate of user k , $d_m(t)$ is the flight vector of UAV m , $v_m(t)$ is the velocity vector of UAV m , and $P_J(t)$ is the jammer's jamming power. Note that the jamming power is estimated by Algorithm 3. The state space can be represented as $S = \{s_t | t = 1, 2, \dots, T\}$. In the state space, when calculating the achievable rate of each UAV-user link, it involves channel state information, which can only be locally observed and is not observable by other UAV-users. Since $R_k(t)$ includes information about other UAVs, this item is difficult to observe. In other words, each agent can only observe the information $\gamma_{m,k}(t)$ connected to its own user. This means that the UAV agent m must rely on a partial observation of the environment to decide its actions. For the UAV agent m , the observation space could be represented as

$$o_t = \{\gamma_{m,k}(t), d_m(t), v_m(t), P_J(t)\}. \quad (30)$$

All the observed results are stored in the experience replay pool $\mathcal{W}$, allowing for efficient utilization of past experiences during training.

Action: The action is the output of the agent based on its current state and is received by the environment. For the UAV m at time slot t , the action space can be represented as

$$a_t = \{\zeta_m(t), v_m(t), P_m(t)\}. \quad (31)$$

where $\zeta_m(t) \in [0, 2\pi]$ represents the flight direction of the UAV. $v_m(t)$ represents the velocity vector of the UAV. $P_m(t)$ represents the transmission power of the UAV. The UAV selects different transmission powers $P_m(t)$ at time slot t to provide services to ground users and determine the next flight direction and velocity.

Transition probability: This represents the probability of the agent transitions to the next state s_{t+1} given the current state of the environment s_t and the action a_t taken by the agent. Formally, it can be denoted as $P(s_{t+1}|a_t, s_t)$.

Reward function: In the optimization problem (22), the goal is to maximize the achievable sum rate of all users in the system while minimizing the power consumption of the all UAVs.

Therefore, the achievable sum rate of all users within a period T is designed as one of the reward functions. To guide the UAV to fly closer to the ground users and farther away from the jammer, additional reward is given based on the Euclidean distance between the UAV and the ground users or jammers. In addition, corresponding penalties should be given when the constraints in (22) are not satisfied. For example, when the UAV's flying range or transmission power limit is exceeded, a penalty function should be considered. Based on the above analysis, the reward function is defined as

$$R_t = \sum_{k=1}^K R_k(t) + R_{dis}^M(t) - R_{dis}^J(t) - e_1 \varepsilon_1(t) - e_2 \varepsilon_2(t), \quad (32)$$

where $\sum_{k=1}^K R_k(t)$ represents the sum of achievable data rates for all users. $R_{dis}^M(t)$ is the reward function for UAV flying closer to the users, $R_{dis}^J(t)$ is the penalty function for UAV flying closer to the jammers, and they are related to the Euclidean distance between the UAV and the users or jammers, which can be expressed as $R_{dis}^M(t) = \frac{1}{MK} \sum_{m=1}^M \sum_{k=1}^K \frac{k_{dis}^M}{\sqrt{(x_m(t)-x_k)^2+(y_m(t)-y_k)^2+h_m(t)^2}}$ and $R_{dis}^J(t) = \frac{1}{M} \sum_{m=1}^M \frac{k_{dis}^J}{\sqrt{(x_m(t)-x_J)^2+(y_m(t)-y_J)^2+h_m(t)^2}}$, where k_{dis}^M and k_{dis}^J are manually set hyperparameters. $e_1 \varepsilon_1(t)$ and $e_2 \varepsilon_2(t)$ correspond to the penalty functions when the UAV's flying range is not satisfied and when the transmission power exceeds the limit, respectively. e_1, e_2 represents the weight coefficient of the constraint conditions. $\varepsilon_1(t), \varepsilon_2(t)$ are binary functions. $\varepsilon_1(t) = 0$ represents satisfying the UAV's flying range, $\varepsilon_1(t) = 1$ represents exceeding the UAV's flying range, $\varepsilon_2(t) = 0$ represents satisfying the UAV's transmission power, and $\varepsilon_2(t) = 1$ represents exceeding the UAV's transmission power.

The agent observes the environment, takes corresponding actions, and obtains reward from the environment. This interactive process can be represented by a sequence $\varsigma = \{s_1, a_1, r_1, \dots, s_T, a_T, r_T\}$. The probability of the sequence ς can be expressed as

$$P(\varsigma) = s_1 \prod_{t=1}^T \pi(a_t, s_t) P(s_{t+1} | a_t, s_t), \quad (33)$$

where $\pi(a_t, s_t) = P(a_t | s_t)$. Therefore, the transition probability can be expressed as

$$P(s_{t+1} | a_t, s_t) = \int_S \Gamma(s_t, a_t, s_{t+1}) ds_{t+1}, \quad (34)$$

where $\Gamma(s_t, a_t, s_{t+1})$ represents the transition function, representing the joint probability of transitioning from state s_t to $s_{t+1} \subseteq \mathcal{S}$ given action a_t .

Each agent has the state value function (SVF) $V_\pi(s_t)$ and state-action value function (SAVF) $Q_\pi(s_t, a_t)$, where π represents the current policy. The SAVF represents the expected long-term cumulative return that an agent can obtain in a given state by taking a certain action. It measures the goodness or badness of taking a specific action in a particular state. The SAVF $Q_\pi(s_t, a_t)$ can be expressed as:

$$Q(s_t, a_t) = E_{s_{t+1} \sim \mathcal{W}} [r(s_t, a_t) + \gamma V_\pi(s_{t+1})]. \quad (35)$$

The SVF represents the expected long-term cumulative return that an agent can obtain in a given state, measuring the goodness or badness of the current state. The SVF $V_\pi(s_t)$ can be expressed as:

$$V(s_t) = E_{a_t \sim \pi} [Q(s_t, a_t) + \alpha H(\pi(\cdot|s_t))], \quad (36)$$

where $H(\pi(\cdot|s_t)) = E[-\log \pi(\cdot|s_t)]$ represents entropy. The higher the entropy, the greater the agent's exploration in the environment, enabling it to find a more efficient strategy, which helps accelerate subsequent policy learning. α represents hyperparameter, which aims to control the randomness of the optimal strategy and balance the importance of entropy relative to reward. The soft actor critic (SAC) algorithm is a model free DRL algorithm based on maximum entropy. It introduces the concept of maximum entropy on top of maximizing future cumulative rewards to enhance robustness and the exploration capability of the agent. The objective of the SAC algorithm is to maximize both future cumulative rewards and entropy, making the policy as random as possible.

The advantage value function (AVF) $A_\pi(s_t, a_t)$ is used to assess the advantage or value of specific actions relative to general actions, and it can be calculated by the difference between the SVF and the SAVF, expressed as:

$$A_\pi(s_t, a_t) = Q_\pi(s_t, a_t) - V_\pi(s_t). \quad (37)$$

The computation result of the AVF $A_\pi(s_t, a_t)$ can help the agent determine the relative value of an action in a specific state. A positive value demonstrates that the current action is more advantageous than the average level, while a negative value is less advantageous. This guides the decision-making and policy updates of the agent. By maximizing the AVF $A_\pi(s_t, a_t)$, the agent can choose relatively better actions and gradually improve its decision-making capability and performance. Based on the above analysis, the objective function is represented as

$$J(\pi) = E_{s_t} \left[P_\pi(s_t) \sum \pi(s_t, a_t) A_\pi(s_t, a_t) \right], \quad (38)$$

where $P_\pi(s_t)$ represents the probability of observing state s_t when selecting policy π .

V. MADRL ALGORITHM WITH CENTRALIZED TRAINING AND DISTRIBUTED EXECUTION

A. Centralized Training and Distributed Execution

Due to the huge state space in the multi-UAV environment, RL based methods struggle to find optimal strategies. The MADRL model is a RL model designed specifically for multi-UAV environments to solve the problem of a large state space and achieve cooperation among agents [43]. The proposed MADRL model is divided into centralized training and distributed execution, as detailedly shown in Fig. 3.

In the centralized training stage, each UAV is treated as an agent and interacts with the environment independently. Each agent stores the experience obtained from interacting with the environment in an experience replay pool, and then uses this pool to train an soft actor-critic network. In the distributed execution stage, each agent interacts with the environment based on the actions output by the actor network and receives a reward. The agents use the critic network to estimate the quality of their

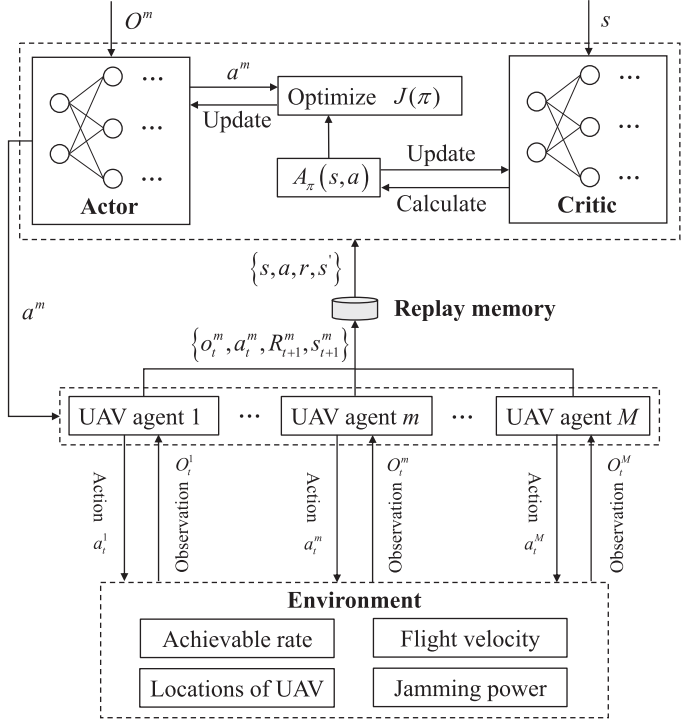


Fig. 3. Illustration of MADRL framework.

actions, and use the temporal difference algorithm to update the parameters of the actor and critic networks.

1) *Centralized Training Process:* The centralized training stage is completed in the HAP. Each UAV needs to learn the next flight position and transmission power. In the training process, the soft actor-critic network is updated by randomly sampling small batches of observations $\{s, a, r, s'\}$ from the experience replay pool to avoid correlations among total observations and improve learning efficiency. This approach effectively utilizes previous experiences, making the model more stable and converging faster. It is worth noting that since the observation results of all UAVs are stored in the experience replay pool, management and control of the experience replay pool are necessary to avoid experience repetition or outdated issues. Additionally, to improve learning efficiency, suitable sampling strategies and optimization algorithms need to be designed to better utilize the experience.

In the actor network, a deep neural network (DNN) is used to parameterize the policy function, and the policy is generated by a Gaussian distribution [44], which can be expressed as $\pi_{\theta_A}(s, a) = \frac{1}{\sqrt{2\pi}\tilde{\sigma}(s)} e^{-\frac{a-\tilde{\mu}(s)}{2\tilde{\sigma}(s)^2}}$. The parameters of the actor DNN are denoted as θ_A . The standard deviation $\tilde{\sigma}(s)$ and mean value $\tilde{\mu}(s)$ are used to generate actions. The critic network calculates the advantage function, which guides the gradient of the actor network towards moving in a low-cost direction.

The training process is conducted offline, running on the HAP. The advantage of using offline centralized training and online distributed execution methods is that it allows all communication and computation overhead to be placed in the training phase. As a result, it can significantly improve the execution speed compared to traditional algorithms during the testing phase.

Algorithm 4: MADRL Based Algorithm for Multi-UAV Network.

Input: The observation space $\mathcal{O}$, the environment of flight velocity, location of UAVs and jamming power, the action space $\mathcal{A}$.

Output: The UAV's flight trajectory and total transmission power.

- 1: Initialize the neural network, experience replay pool and parameter settings for the air-ground integrated network.
- 2: **for** each time slot $t = 1, \dots, T$ **do**
- 3: **for** each UAV agent $m = 1, \dots, M$ **do**
- 4: Observe the environment and obtain the observation o_t^m and select an action a_t^m by importance sampling.
- 5: **end for**
- 6: Evaluate the reward function to obtain the reward R_t^m and transition to a new state s_{t+1} .
- 7: All agents take actions according to the reward and interact with the environment to obtain the next reward R_{t+1}^m at the next time.
- 8: **for** each UAV agent $m = 1, \dots, M$ **do**
- 9: Compute R_{t+1}^m and store $\{o_t^m, a_t^m, R_{t+1}^m, s_{t+1}^m\}$ in the experience replay pool.
- 10: **end for**
- 11: **end for**
- 12: **for** each UAV agent $m = 1, \dots, M$ **do**
- 13: Use **Algorithm 5** to train the soft actor-critic DNN.
- 14: **end for**

2) *Distributed Execution Process:* In the distributed execution process, each UAV outputs action sequences by the trained actor DNN to adjust the flight position and transmission power for the next time. It is important to note that since each UAV independently uses the actor DNN to generate action sequences, if actions are updated simultaneously, each UAV is unaware of the actions chosen by other UAVs. Therefore, to coordinate the actions of multiple UAVs, a critic DNN is used to evaluate the actions of each UAV and update the parameters of both the actor DNN and critic DNN based on the temporal difference algorithm. This distributed execution mode enables multiple UAVs to collaborate and achieve joint services while avoiding interference between each other.

Based on the analysis above, we propose the MADRL based algorithm for multi-UAV network as shown in Algorithm 4. Firstly, the DNN, experience replay pool $\mathcal{W}$, and parameter settings are initialized. In each training episode time slot t , the agent observes the state o_t^m by importance sampling to obtain the achievable rate, UAV location, flight velocity, and jamming power. This serves as an approximate method to replace sampling, with only the actor network being active at this stage. Subsequently, the state sequence is input into the actor DNN to determine the action that maximizes the received reward R_{t+1}^m . Lastly, the experience replay pool stores the transition tuple $\{o_t^m, a_t^m, R_{t+1}^m, s_{t+1}^m\}$ for each agent.

B. Training With Improved PPO Algorithm

Note that in action (31), there exist continuous variables $\{\zeta_m(t), v_m(t), P_m(t)\}$. Although policy gradient methods, such

as deep deterministic policy gradient (DDPG), can effectively obtain policies for continuous space, they suffer from convergence issues when dealing with high-dimensional continuous action space and persistent task. In comparison to DDPG, PPO adopts a policy optimization approach that limits the policy update step size, ensuring only moderate changes in the policy with each update, thus improving the sampling efficiency and stability of traditional policy gradient algorithms. Though trust region policy optimization (TRPO) can effectively handle action space with continuous variables, TRPO's second-order optimization is insufficient and directly using it to train the network is extremely time-consuming [45]. Additionally, TRPO requires adjustment of multiple parameters, such as maximum step size and thresholds. In contrast to TRPO, PPO only requires control of the update step size of hyperparameters, making it simpler and easier to use in implementation, while still providing performance close to TRPO. Therefore, this paper proposes a PPO algorithm based on clip and count to train the soft actor-critic network, which simplifies the equation using a clipping ratio function.

Firstly, the probability ratio between the new policy π_θ and old policy $\pi_{\theta_{old}}$ is defined as $\Upsilon(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$. To train this network, a loss function needs to be defined to guide the update of parameters. The loss function can be defined as

$$L(s_t, a_t, \theta_{old}, \theta)$$

$$= \min \left[\Upsilon(\theta) A_{\pi_{\theta_{old}}}(s_t, a_t), \text{clip} \left(\frac{\Upsilon(\theta)}{1 - \varepsilon}, \frac{\Upsilon(\theta)}{1 + \varepsilon} \right) A_{\pi_{\theta_{old}}}(s_t, a_t) \right], \quad (39)$$

where $\text{clip}(\cdot)$ is the clipping function, $A_{\pi_{\theta_{old}}}(s_t, a_t)$ is the AVF, and ε is a clip parameter used to limit the range of $\Upsilon(\theta)$. The update method for the policy parameters is as follows:

$$\theta_{m+1} = \arg \max_{\theta} \frac{1}{|\mathcal{W}_m| T} \sum_{\tau \in \mathcal{W}_m} \sum_{t=0}^T L(s_t, a_t, \theta_{old}, \theta). \quad (40)$$

However, the traditional PPO algorithm might exhibit limited flexibility since the clipping parameter ε is fixed [26]. To solve this problem, we assume that ε follows a normal distribution $\varepsilon \sim N(\hat{\mu}, \hat{\sigma}^2)$. The modification means that at the beginning of training, due to the differences between the old and current policies, the agent is subject to a larger range of constraints. As training progresses, the differences gradually decrease, and therefore, a smaller range of constraints should be implied to the agent to accelerate algorithm convergence.

Lastly, we apply PPO to the network architecture of both the policy and value function. The objective of updating the critic network is to minimize the loss function, which can be expressed as

$$L(s_t, a_t, \phi_{old}, \phi) = \left(V_{\phi_{old}}(s_t, a_t) - \hat{R}_t \right)^2, \quad (41)$$

where $\hat{R}_t$ is defined as $\hat{R}_t = \gamma^{t-1} r_t$. ϕ represents the value function parameters, which can be updated by

$$\phi_{m+1} = \arg \min_{\phi} \frac{1}{|\mathcal{W}_m| T} \sum_{\tau \in \mathcal{W}_m} \sum_{t=0}^T L(s_t, a_t, \phi_{old}, \phi). \quad (42)$$

Algorithm 5: Improved PPO Algorithm Based on Clip and Count.

Initialize: Policy parameters θ_1 of the actor network and value function parameters ϕ_1 of the critic network.

- 1: **for** $m = 1, 2, \dots$ **do**
- 2: Collect the trajectory set based on the running policy $\pi_m = \pi(\theta_m)$.
- 3: Calculate the reward $\hat{R}_t$.
- 4: Calculate the SVF, the SAVF, and the AVF based on formulas (35), (36), and (37), respectively.
- 5: Adjust the target entropy to balance exploration and exploitation. If the entropy of the policy is lower than the target entropy, increase the value of the target entropy to encourage more exploration.
- 6: Set the clipping parameter ε according to the distribution $\varepsilon \sim N(\hat{\mu}, \hat{\sigma}^2)$.
- 7: Update the policy parameters θ_{m+1} of the actor network based on formulas (39) and (40).
- 8: Update the value function parameters ϕ_{m+1} of the critic network based on formulas (41) and (42).
- 9: **end for**

We train the soft actor-critic network by the improved PPO algorithm based on clipping and counting, as shown in detail in Algorithm 5. Firstly, initialize the policy parameters θ_1 and value function parameters ϕ_1 . Then collect the trajectory set based on the policy π_m and calculate the reward $\hat{R}_t$. Calculate the SVF, the SAVF, and the AVF based on formulas (35), (36), and (37), respectively. Then set the clipping parameter ε according to the distribution $\varepsilon \sim N(\hat{\mu}, \hat{\sigma}^2)$ to train the soft actor-critic network. Lastly, update the new policy parameters θ_{m+1} and the value function parameters ϕ_{m+1} of the soft actor-critic network based on formulas (39)–(42).

C. Complexity Analysis

This subsection provides a complexity analysis of the MADRL algorithm. The algorithm consists of two main components within one period T : Centralized training and distributed execution processes. During the centralized training process, each UAV agent uses an actor DNN to output its actions, taking importance sampling as input. Therefore, the time complexity of centralized training can be represented as $O(\sum_{i=1}^I N_i N_{i-1})$, where $N_i, i = 1, 2, \dots, I$ is the number of neurons in the fully connected layers of the actor DNN [46]. During the distributed execution process, executing actions for each UAV is the most computationally demanding task. Referring to (31) and (38), the computational complexity for action selection and estimating the SAVF amounts to $O(MKT)$.

Overall, the complexity analysis suggests that the training complexity of MADRL algorithm is influenced by the number of neurons in the actor DNN, while the execution complexity depends on the number of UAVs and users involved. The total complexity of the algorithm is $O(\sum_{i=1}^I N_i N_{i-1} + MKT)$.

VI. SIMULATION RESULT AND ANALYSIS

The section presents simulation results and analysis to validate the effectiveness of the proposed anti-jamming algorithm.

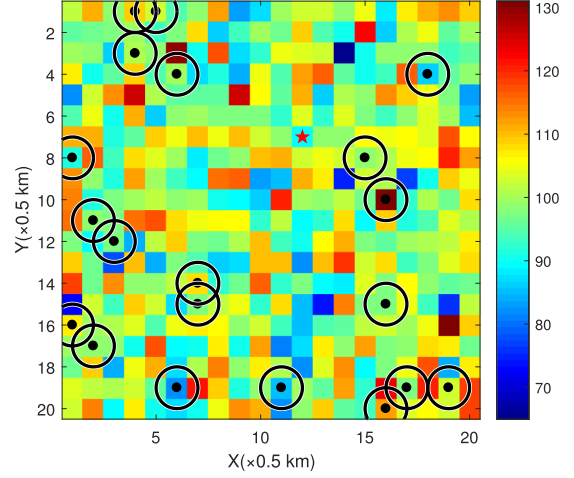


Fig. 4. The initial deployment positions of the UAVs.

The simulation environment is set up with 1 ground jammer, 3 air jammers, 1 HAP, 20 UAVs, and 20 users in an integrated air-ground network. All users are randomly distributed within an area of $10\text{km} \times 10\text{km}$ square meters and the discrete grid count is $X = 400$. UAV data collection radius is 0.5km . UAV flight speed $v_m(t)$ is 6m/s – 20m/s . The altitude of the UAVs, air jammers, and HAP are $3 \sim 7\text{ km}$, $5 \sim 7\text{ km}$, and 10km . Some HAP related parameters are set as $G_{uh} = 15\text{dB}$, $k_B = 1.38 \times 10^{-23}\text{ J/K}$, $T_s = 1000\text{K}$ [29]. The jamming signal strength level N_{num} is 20. Each agent's actor and critic networks consist of two fully connected hidden layers with 50 neurons each. The learning rate for both actor and critic networks is set to 0.0001. Additionally, the discount factor is set to 0.999. The clipping parameter distribution is $\varepsilon \sim N(1, 0.3)$, which follows a normal distribution [44]. The weight coefficient (e_1, e_2) is $(50, 500)$ and (w_1, w_2) is $(10, 10)$.

Firstly, we simulate the performance of the proposed UAV deployment and networking algorithm for uplink process. Fig. 4 shows the initial deployment positions of the UAVs. In the initial deployment, the UAVs are distributed at various grid nodes, with some high-data-traffic hotspots not adequately covered. There is overlap in the data collection ranges of some UAVs, which reduces their data collection efficiency as the overlap area increases. Additionally, some UAVs are within the interference range of interfering devices, preventing them from collecting data and further reducing data collection efficiency. Therefore, it is necessary to redeploy the UAV positions to avoid overlap in coverage areas and to move out of interference ranges. Each UAV optimizes its position according to the node deployment algorithm, and the results are shown in Fig. 5. Through the congestion game of the UAV group, each UAV adjusts its position to avoid grid areas with interfering devices and move towards hotspots with more data traffic. At the same time, UAVs try to avoid overlapping data collection grids with each other, thus improving the overall data collection efficiency of the entire UAV group.

The UAV swarm, air jammers, and HAP undergo dynamic networking through alliance games, as shown in the final outcome. Fig. 6 illustrates the formation of a three-dimensional alliance,

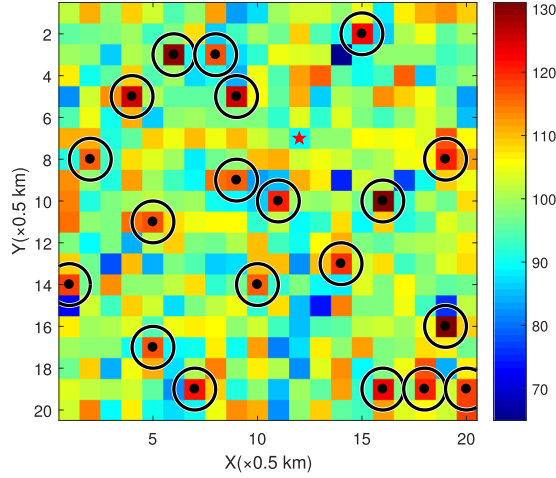


Fig. 5. The optimized deployment positions of the UAVs.

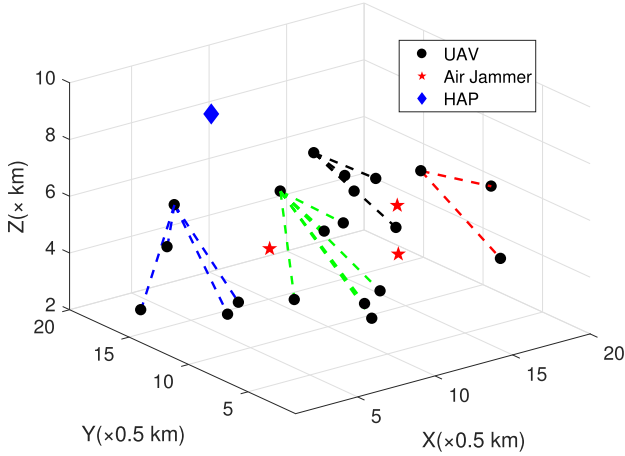


Fig. 6. The UAV alliance clusters in 3D vision.

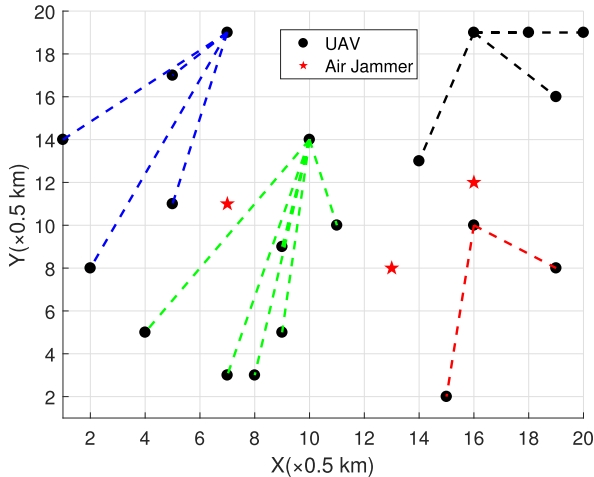


Fig. 7. The UAV alliance clusters in 2D vision.

while Fig. 7 depicts a two-dimensional plane. Lines of the same color represent the same alliance group. Within the same alliance group, UAVs first select cluster head UAVs to aggregate data, and then the cluster head UAVs upload the data to the HAP to complete the data uploading task.

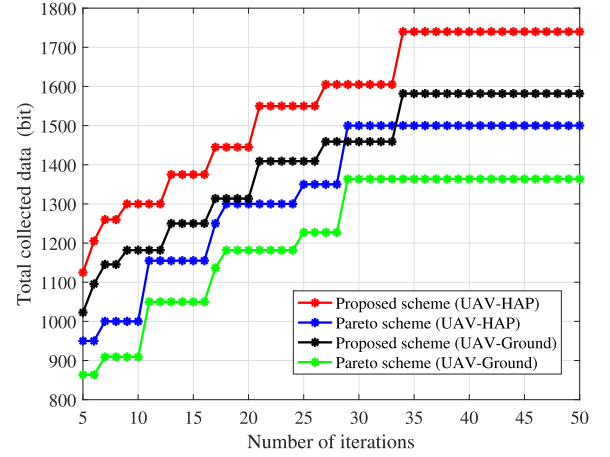


Fig. 8. The total data amount upload.

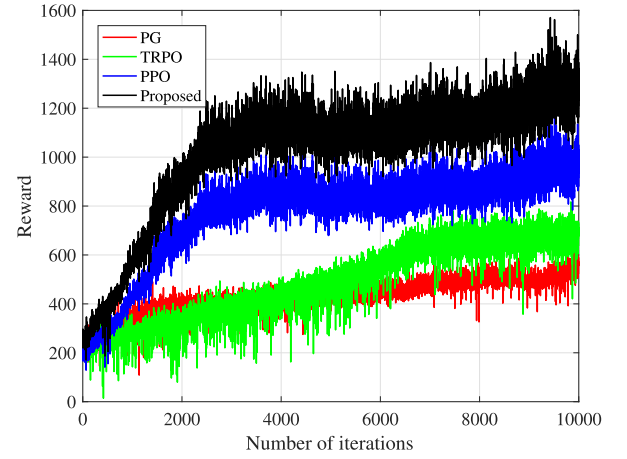


Fig. 9. System reward with different algorithms.

Fig. 8 illustrates the total data upload amount for our proposed alliance game grouping algorithm and Pareto scheme [48]. When the algorithm converges, the proposed algorithm achieves approximately 17% increase in total data upload compared to the Pareto scheme, thereby verifying the anti-jamming effectiveness of the proposed UAV deployment and networking algorithm. In addition, The UAV-HAP communication exceeds the total amount of data uploaded on the UAV-ground server communication. This is because the UAV-ground server communication may be obstructed. The obstructed UAVs may not be able to upload data to the ground server, resulting in a decrease in the total amount uploaded. This validates the benefits of HAP compared to ground base stations.

Next, we simulate the performance of the proposed MADRL algorithm for downlink process. We compared it with traditional PPO algorithm [45], policy gradient (PG) method [49], and TRPO algorithm [50]. We select one of the UAV alliance clusters and apply the MADRL algorithm for simulation.

Fig. 9 compares the system rewards obtained by the three algorithms mentioned above. The simulation results clearly demonstrate that the algorithm proposed in this paper possesses two significant advantages over several other algorithms. Firstly, the agent is capable of attaining a higher number of the reward.

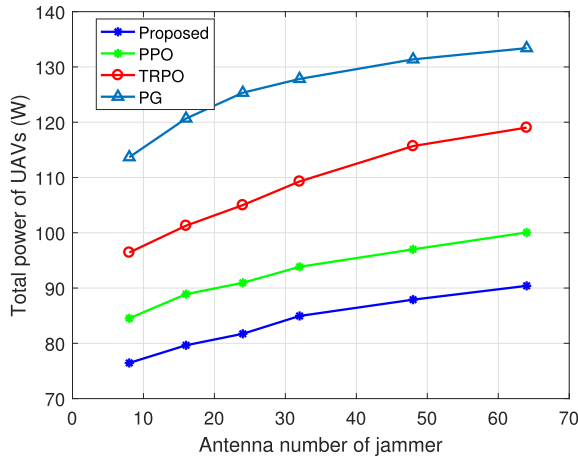


Fig. 10. Total power of UAVs with different antenna number of jammer.

When all four algorithms reach a converged equilibrium, the proposed algorithm obtains an average reward value of nearly 1200, while traditional PPO, TRPO, and PG algorithms only obtain reward values of approximately 900, 710, and 570, respectively. This implies that the proposed algorithm can achieve superior policies. Secondly, it exhibits a notably expedited convergence speed, typically converging after approximately 3500 iterations, while traditional PPO, TRPO, and PG algorithms tend to converge after about 4000, 7000, and 6000 iterations, respectively. This compellingly suggests that the proposed improved algorithm can effectively facilitate network training. This is because the improved PPO algorithm uses a clipping ratio variable, which limits the magnitude of policy updates and prevents excessive changes in the policy. It ensures that the training process is stable and does not overadjust the policy, resulting in higher rewards and faster convergence speed compared to the other algorithms.

Fig. 10 compares the total power consumption of UAVs for anti-jamming transmission under different antenna number of the jammer, based on the results after the algorithms have converged. The antenna number of the ground jammer is set as [8, 16, 24, 32, 48, 64]. Simulation result indicates that the total power consumption increases with the antenna number of jammer. This is due to the increase in the number of jamming antennas, the jamming capability to users has also been enhanced, thus requiring more transmission power consumption. However, the total power of the proposed algorithm is less than that of the other algorithms under the same antenna number condition. Since the traditional PPO algorithm performs the best among the three compared algorithms, we use the traditional PPO algorithm and the proposed algorithm as examples for illustration. When the number of jammer antennas is set to 32, the proposed algorithm consumes a total UAV power that is 8.9W lower than that of the traditional PPO algorithm, and 42.8W lower than that of the PG algorithm. The proposed algorithm could reduce the power consumption and burden of UAVs, thereby decreasing system overhead.

Fig. 11 compares the system achievable sum rate under different jamming power, based on the results after the algorithms have converged. The jamming power is set as [40, 60, 80, 100] W. The simulation results demonstrate that as the jamming power increases, the system achievable sum rate decreases. The

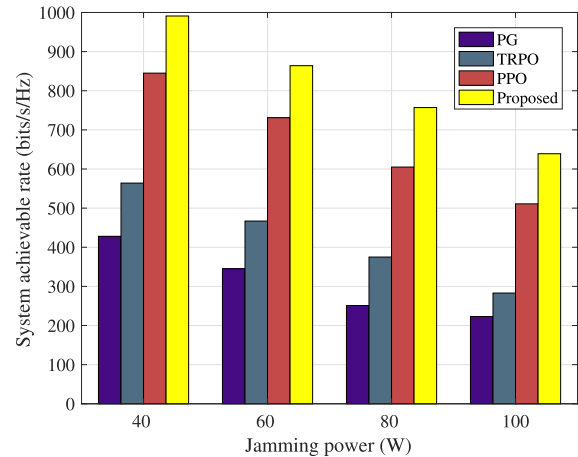


Fig. 11. System achievable sum rate with different jamming power.

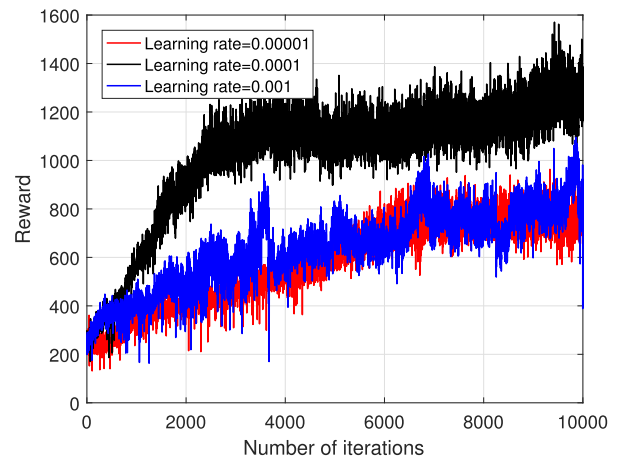


Fig. 12. System reward with different learning rate.

enhancement of jamming power leads to an increase in the jamming effect on the system, thereby reducing the achievable sum rate of the system. This phenomenon is in line with objective principles. However, our proposed algorithm outperforms other algorithms in terms of achievable sum rate under the same jamming power condition. Taking the traditional PPO algorithm and the proposed algorithm as examples, when the jamming power is set to 60W, the proposed algorithm achieves a system achievable sum rate that is 12.4% higher than that of the traditional PPO algorithm. Since the reward function is designed as one of the indicators based on the achievable sum rate, obtaining higher reward means a larger achievable sum rate. Therefore, the algorithm proposed in this paper can achieve the highest reward, indicating that the achievable sum rate is also the largest. This observation validates that our proposed algorithm effectively enhances the regional coverage capability of the communication system in the presence of jamming.

Fig. 12 simulates the performance by setting different learning rate δ . The learning rate is set to three values [0.001, 0.0001, 0.00001]. The simulation results show that when there are fewer iterations, the reward of a higher learning rate, such as $\delta = 0.001$, is lower than that of a lower learning rate, such as $\delta = 0.0001$. However, as the number of iterations increases, the reward also increases and eventually converges. This

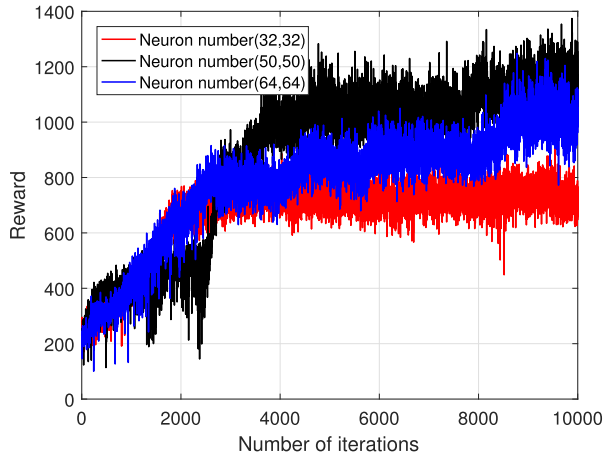


Fig. 13. System reward with different neurons layers.

is due to the fact that a larger learning rate, such as $\delta = 0.001$, may cause fluctuations in the training process, making it struggle to obtain the optimal strategy, while a smaller learning rate, such as $\delta = 0.00001$, can lead to excessively long training time. In a word, when the learning rate is too small, both the convergence speed and the obtained reward are low. When the learning rate is too large, the algorithm is difficult to converge. Therefore, choosing a moderate learning rate allows for obtaining higher rewards while ensuring fast convergence of the algorithm. We set the learning rate to $\delta = 0.0001$ with practical execution consideration of the algorithm.

The proposed algorithm's performance under varying numbers of neurons is simulated in Fig. 13. To assess the impact of different numbers of neurons in the DNN structure on algorithm performance, the number of neurons in the two hidden layers are set as (32,32), (50,50), and (64,64). The results indicate that increasing the number of neurons from (32,32) to (50,50) improves the obtained reward but reduce the convergence speed of the algorithm. However, setting a large number of neurons, such as (64,64), results in reduced reward. The reason for this is that a high number of neurons can lead to overfitting of the DNN, ultimately reducing the reward. A smaller number of neurons can achieve a faster convergence speed, while a larger number of neurons may lead to significant reward fluctuations and difficulty in convergence. It is advisable to select a moderate number of neurons, which allows for obtaining higher rewards while ensuring fast algorithm convergence. Hence, this paper adopts a DNN structure with (50,50) neurons to prioritize performance while minimizing computational burden.

To verify the performance of dynamically jamming strategy, we change the jamming strategy and then simulate until convergence. As shown in the Fig. 14, the proposed algorithm converges after nearly 4000 iterations. At the iterations of 10000, the first instance of a new unforeseen external jamming occurs, causing the proposed algorithm to reiterate until convergence. However, the convergence speed is relatively slow compared to before. The trained model is placed on the HAP. When a second instance of unforeseen external jamming occurs, the UAV downloads the trained model from the HAP, thus accelerating

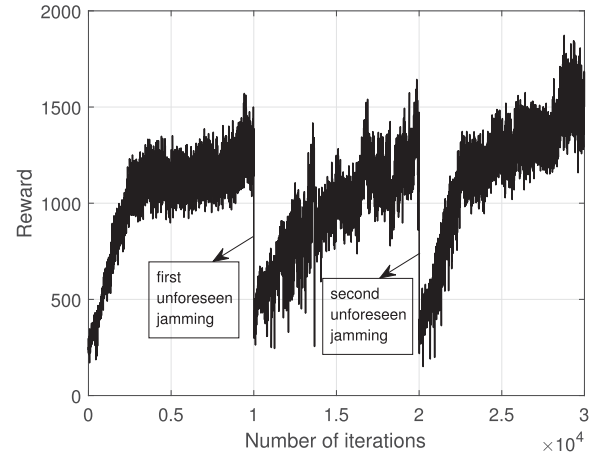


Fig. 14. Reward with dynamically jamming strategy.

convergence speed and obtaining a higher reward compared to the first jamming instance.

VII. CONCLUSION

This paper proposed an anti-jamming transmission algorithm based on game theory and MADRL for the integrated air-ground network. For uplink process, we propose UAV deployment schemes based on congestion game model and dynamic networking schemes based on coalition game, aiming to defend malicious jamming from ground and air jammers, effectively enhancing the anti-jamming transmission capability of air-ground networks. For downlink process, the joint trajectory and power optimization problem was transformed into a POMDP problem. The proposed approach followed a centralized training and distributed execution framework. During the centralized training process, experiences of each agent interacting with the environment were stored in an experience replay pool and then used to train the soft actor-critic network. During the distributed execution process, each UAV outputs actions based on observations and adjusted its flight position and transmission power for joint service provision. An improved PPO algorithm was proposed to update the soft actor-critic network parameters. Simulation results demonstrate that the proposed method can achieve higher system achievable sum rate, lower power consumption, and faster convergence speed compared to traditional algorithms.

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